Full Mark is 45.

Score:_____

Mark Scheme - G1 Vocabulary Quiz – 1

1.

1) $\overrightarrow{OH}$ B1

$\mathbf{a}+\mathbf{b}$ B1

$\mathbf{2}$) Collinear B1

2.

Domain is the set of the possible ___*input or x values*_____ of a function. [1]

Range is the set of the possible __*output or y values*_____ of a function. [1]

3.

List the domain and range of the following function. Then list its inverse domain and range. And write down the features of the graphs between f and f^{-1}.

$$f(x)=\sqrt{x-2}$$

Domain:_____$x \geq 2$_____[1] **must be in terms of x. Do not accept y or f.**

Range:_____$f \geq 0$_____ [1] **must be in terms of y or f. Do not accept x.**

For the inverse function, $f^{-1}(x)=x^2+2$,

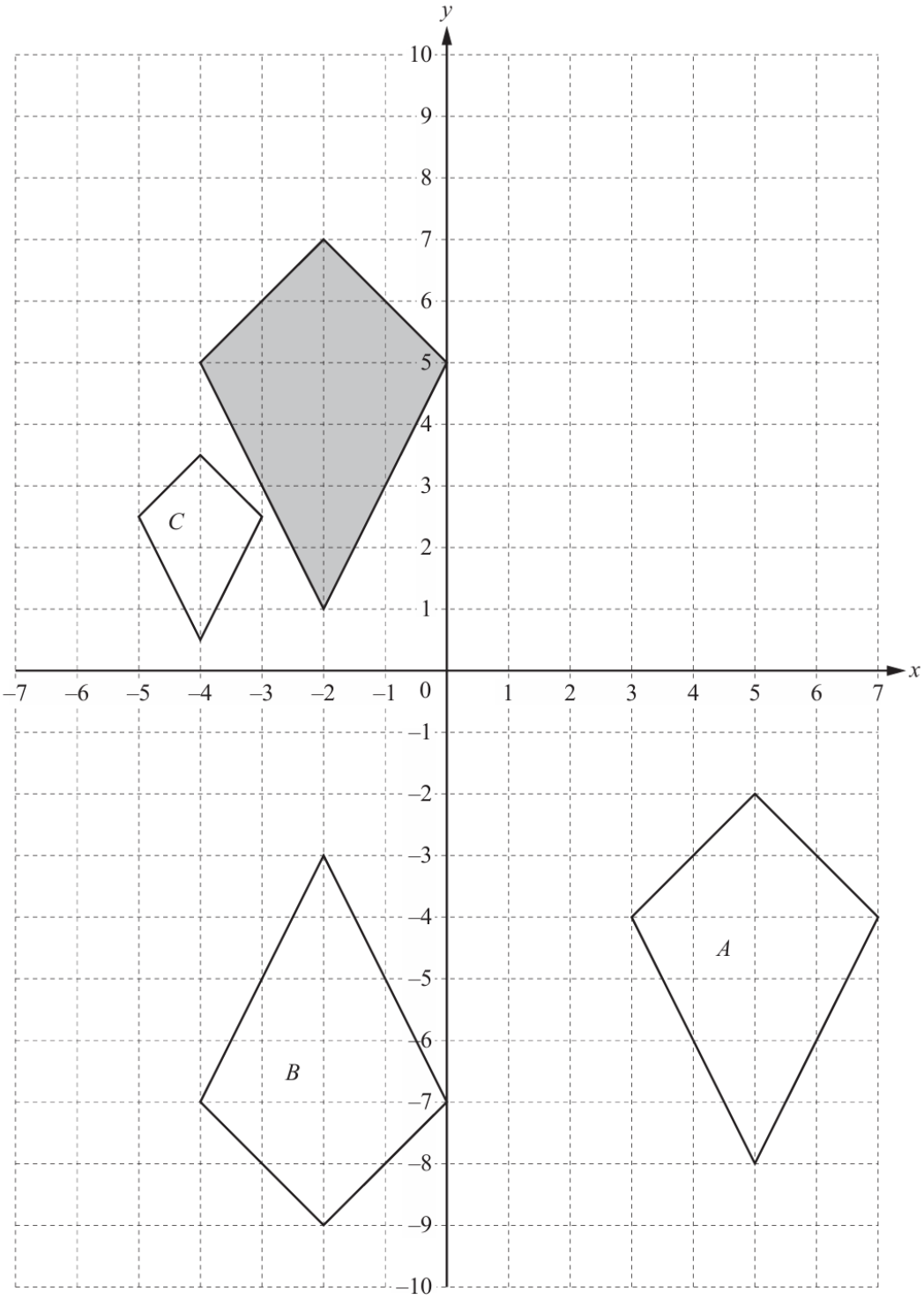
Domain:_____ $x \geq 0$_____[1] **Follow Through their original range, must be in terms of x. Do not accept y or f.**

Range:_____ $f^{-1}(x) \geq 2$_ _[1] **Follow Through their original domain, must be in terms of y or f^{-1}. Do not accept x.**

The graph of f and f^{-1} are ____reflection/symmetrical to the line y=x_____. [2]

B1 B1

4.



Describe fully the **single** transformation that maps the shaded quadrilateral onto

(i) quadrilateral A,

Answer(c)(i) .....

..... [2]

(ii) quadrilateral B,

Answer(c)(ii) .....

..... [2]

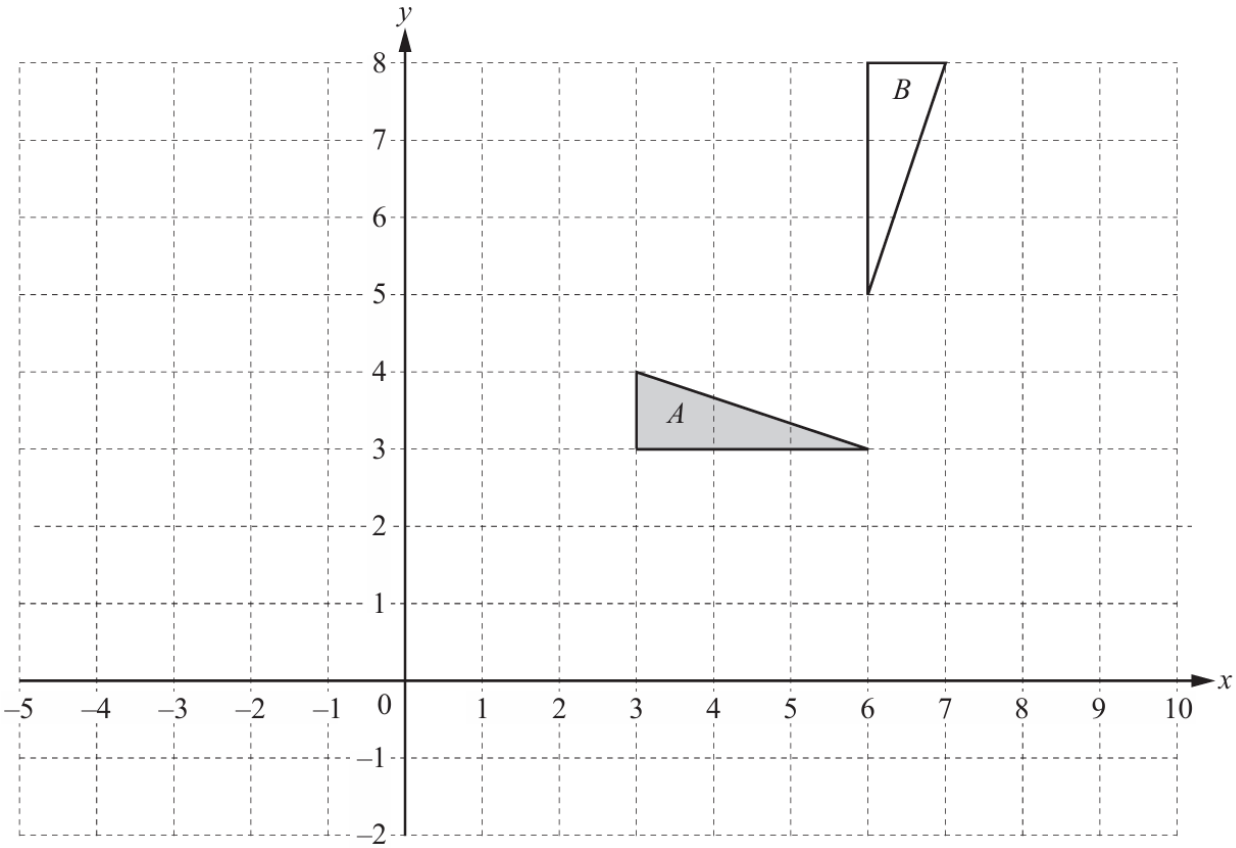
(iii) quadrilateral C.

Answer(c)(iii) .....

..... [3]

(i)	Translation	**1**
	$\begin{pmatrix} 7 \\ -9 \end{pmatrix}$	**1**
(ii)	Reflection	**1**
	$y = -1$ oe	**1**
(iii)	Enlargement	**1**
	[Scale Factor] $\frac{1}{2}$	**1**
	[Centre] $(-6, 0)$	**1**

5.



Describe fully the **single** transformation that maps triangle *A* onto triangle *B*.

.....

..... [3]

Rotation 90º clockwise oe (7, 4)	**1** **1** **1**	Accept –90°

6.

Describe fully the sequence of transformations from $f(x)\ to\ \ -\boldsymbol{f(x+1)}$ [4]

Translation $\binom{-1}{0}$, then followed by reflection, in x axis.

B1 B1 B1 B1

Order can be reversed.

7.

Describe fully the sequence of transformations from $f(x)\ to\ f(4x-3)$ [5]

B1 B1 B1 B1 B1

Translation $\binom{3}{0}$, then followed by stretch, in the x direction, $sf = \frac{1}{4}$

Or stretch, in the x direction, $sf = \frac{1}{4}$, *then followed by Translation* $\begin{pmatrix} \frac{3}{4} \\ 0 \end{pmatrix}$.

8.

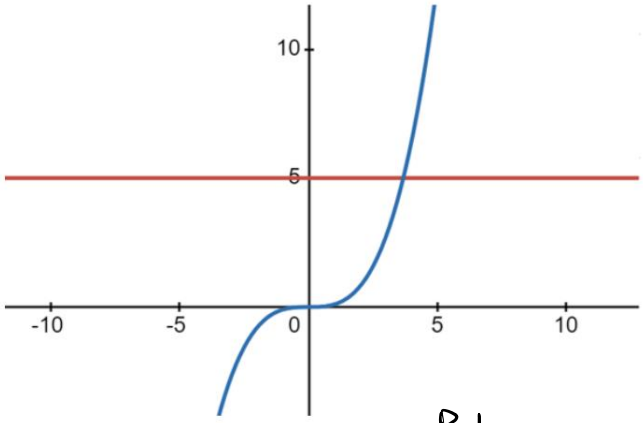
Describe fully the sequence of transformations from $f(x)\ to\ \ \boldsymbol{2f(x)+3}$ [5]

B1 B1 B1 B1 B1

Stretch, in the y direction, $sf = 2$, then followed by Translation $\binom{0}{3}$

Or Translation $\begin{pmatrix} 0 \\ \frac{3}{2} \end{pmatrix}$, *then followed by Stretch, in the y direction,* $sf = 2$

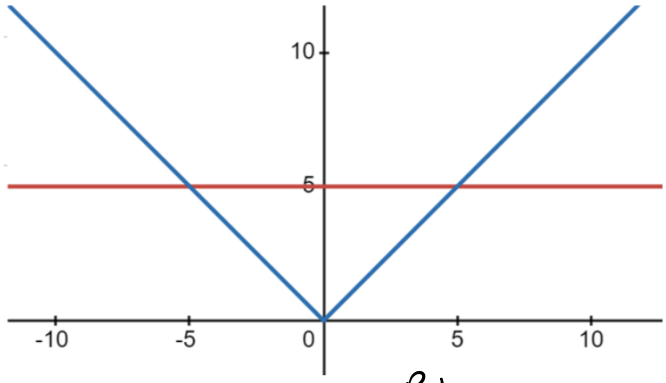
9.



B1 B1

The function above is _____***one-one***_____ mapping type. And it ____***can***___ (can or cannot) be formed an inverse function. [2]

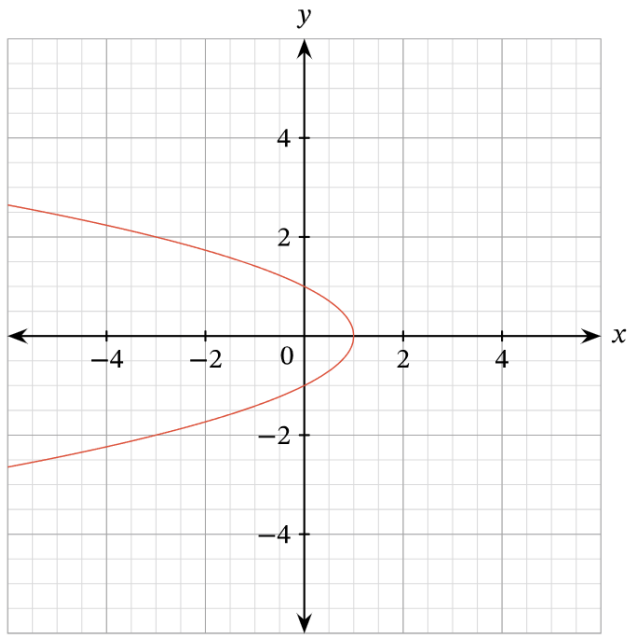
10.



B1 B1

The function above is _____***many-one***_____mapping type. And it ____***cannot***___ (can or cannot) be formed an inverse function. [2]

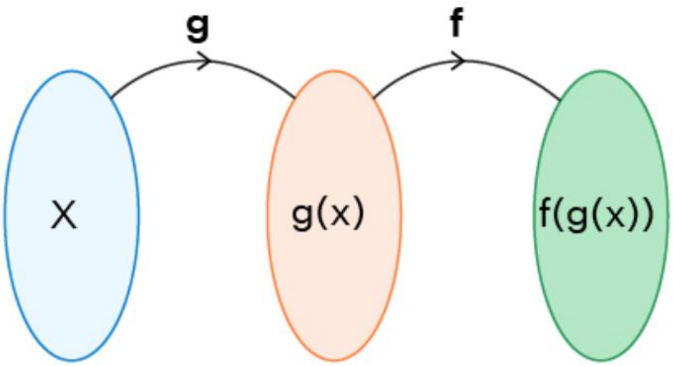
11.



The curve above is ___***one-many***_____ mapping type and it _____***is not***____ (is or is not) a function. [2]

B1 B1

12.



B1

Write down the condition to form a composite function fg: ____***the range of g(x) should be within the domain of f(x)***_____ [2]

B1

13.

Function $f(x) = \frac{1}{x}$ is a self-inverse function. Write down two properties of a self-inverse function.

1) The graph of a self-inverse function is _____***symmetrical to the line y=x***_____[1]
2) If ____$ff(x)$____ $= x$, then it is a self-inverse function. [1]